

Data Wrangling 🤠

While Loops

Lesson 7

while Loops

- Repeat a block of code **as long as** a condition is `True`

```
while [boolean expression]:  
    [code to repeat while True]
```

- Like an `if` statement, but after the block finishes it **jumps back** to the test
- If the test is `False` from the start, the block is skipped entirely

Demo: **while** Loop

```
# loops.py
i = 0

while i < 5:
    print("In repeat block!")
    print(f"i is {i}")
    i = i + 1

print(f"All done! i is now {i}")
```

Run the program with the command: `python -m lessons.loops`

Execution Flow

1. Evaluate the boolean test
2. If `True` → execute the repeat block, then go back to **step 1**
3. If `False` → skip the block and continue

Demo: Infinite Loops

- If the test **never** becomes `False` → the loop runs forever

```
# oops.py
i = 0
while i < 10:
    print("oops")
    # forgot i = i + 1
    # i = i - 1 doesn't ever make the test False
```

- Press `Control + C` to interrupt and stop your program

Demo: Coin Flip Simulation

- Simulation is a very common tool in data science

```
# flip.py
from random import randint
side_flipped = "heads"
heads_count = 0
while side_flipped != "tails" :
    coin = randint(0, 1)
    if coin == 1:
        side_flipped = "tails"
    else:
        side_flipped = "heads"
        heads_count += 1
print(f"We saw {heads_count} consecutive heads!")
```